Emergency Training Simulator POC

Markdown documentation PDF copy

WebSocket error codes

Source: docs/security/websocket-error-codes.md

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Boundary: this project is still a POC, not operational guidance.

WebSocket error codes

The WebSocket endpoint returns simple error messages when a client sends something the backend cannot accept.

The shape is:

```
{
"message_type": "error",
"code": "sequence_number_out_of_order",
"message": "Expected sequence_number 1"
}
```

The code is for software. The message is for a human reading the demo output.

Current codes

| Code | Meaning | | --- | --- | | invalid_initial_message | First message was not start_session or trainee_action. | | invalid_message | Message shape failed Pydantic validation. | | unsupported_message | Message type is not supported in the current WebSocket loop. | | invalid_session_id | Session ID has unsafe characters or is too long. | | invalid_scenario_id | Scenario ID has unsafe characters or is too long. | | session_already_exists | A session with that ID already exists in this backend process. | | active_session_limit_reached | Too many active sessions are already open. | | session_already_ended | The client tried to continue after the scenario ended. | | session_action_limit_reached | The session has already accepted the max number of actions. | | session_id_mismatch | URL, envelope, and event session IDs do not match. | | scenario_id_mismatch | Message scenario ID does not match the active scenario. | | sequence_number_mismatch | Envelope and event sequence numbers do not match each other. | | sequence_number_out_of_order | The client skipped or repeated the next expected sequence number. | | duplicate_event_id | The event ID was already used in this session. | | correlation_event_id_mismatch | Correlation ID does not match the trainee action event ID. | | unknown_session | The backend does not know that session ID. | | session_error | Fallback for an unexpected session-level problem. |

Important note

These errors are local POC guardrails.

They do not replace authentication, authorization, TLS, production rate limiting, or deployment security.